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Self-centering ceramic washer

Abstract

A self-centering ceramic washer is positioned between a feedthrough and a filter circuit board. The washer has openings through which first and second terminal pins extend. A first opening has an inner arcuate portion contacting the first terminal pin and an outer perimeter portion exposing the braze sealing the terminal pin to the insulator. A second opening has an inner arcuate portion contacting the second terminal pin and an outer perimeter portion exposing the braze sealing the terminal pin to the insulator. In an imaginary configuration with the first and second washer openings superimposed one on top of the other, the cumulative arcuate distance of the inner arcuate portions about one of the terminal pins, subtracting overlap, results in a gap between the superimposed washer openings that is less than a diameter of the first and second terminal pins so that the washer is prevented from lateral movement.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3137808	12/1963	Coda et al.	N/A	N/A
3493026	12/1969	Stotler	411/984	F16B 41/002
4793029	12/1987	Burke	411/375	A44B 1/08
5333095	12/1993	Stevenson et al.	N/A	N/A
5489180	12/1995	Ichihara et al.	N/A	N/A
5751539	12/1997	Stevenson et al.	N/A	N/A
5896267	12/1998	Hittman et al.	N/A	N/A
5905627	12/1998	Brendel et al.	N/A	N/A
5973906	12/1998	Stevenson et al.	N/A	N/A
5978204	12/1998	Stevenson	N/A	N/A
5988966	12/1998	Chen	411/372	F16B 37/14
6008980	12/1998	Stevenson et al.	N/A	N/A
6529103	12/2002	Brendel et al.	N/A	N/A
6765779	12/2003	Stevenson et al.	N/A	N/A
6765780	12/2003	Brendel et al.	N/A	N/A
6882248	12/2004	Stevenson et al.	N/A	N/A
7797048	12/2009	Stevenson et al.	N/A	N/A
7878747	12/2010	Dean et al.	N/A	N/A
7957806	12/2010	Stevenson et al.	N/A	N/A
8047753	12/2010	Dean	411/903	F16B 37/14
8095224	12/2011	Truex et al.	N/A	N/A
8179658	12/2011	Stevenson et al.	N/A	N/A

8195295	12/2011	Stevenson et al.	N/A	N/A
8196295	12/2011	Imafuku et al.	N/A	N/A
8321032	12/2011	Frysz et al.	N/A	N/A
8433410	12/2012	Dabney et al.	N/A	N/A
8437865	12/2012	Dabney et al.	N/A	N/A
8483640	12/2012	Stevenson et al.	N/A	N/A
8604341	12/2012	Barry et al.	N/A	N/A
8653384	12/2013	Tang et al.	N/A	N/A
8670841	12/2013	Dabney et al.	N/A	N/A
8712544	12/2013	Dabney et al.	N/A	N/A
8761895	12/2013	Stevenson et al.	N/A	N/A
8855768	12/2013	Dabney et al.	N/A	N/A
8868189	12/2013	Stevenson et al.	N/A	N/A
8918189	12/2013	Dabney et al.	N/A	N/A
8927862	12/2014	Barry et al.	N/A	N/A
8938309	12/2014	Marzano et al.	N/A	N/A
8996126	12/2014	Stevenson et al.	N/A	N/A
9014808	12/2014	Dabney et al.	N/A	N/A
9031670	12/2014	Dabney et al.	N/A	N/A
9064640	12/2014	Brendel et al.	N/A	N/A
9071221	12/2014	Stevenson et al.	N/A	N/A
9233253	12/2015	Stevenson et al.	N/A	N/A
9352150	12/2015	Stevenson et al.	N/A	N/A
9427596	12/2015	Brendel et al.	N/A	N/A
9447810	12/2015	Dean	N/A	F16B 37/14
9463329	12/2015	Frysz et al.	N/A	N/A
9492659	12/2015	Brendel et al.	N/A	N/A
9511220	12/2015	Marzano et al.	N/A	N/A
9521744	12/2015	Barry et al.	N/A	N/A
9757558	12/2016	Stevenson et al.	N/A	N/A
9764129	12/2016	Stevenson et al.	N/A	N/A
9889306	12/2017	Stevenson et al.	N/A	N/A
9895534	12/2017	Stevenson et al.	N/A	N/A
9931514	12/2017	Frysz et al.	N/A	N/A
9993650	12/2017	Seitz et al.	N/A	N/A
10016595	12/2017	Stevenson et al.	N/A	N/A
10016596	12/2017	Stevenson et al.	N/A	N/A
10046166	12/2017	Stevenson et al.	N/A	N/A
10080889	12/2017	Marzano et al.	N/A	N/A
10092749	12/2017	Stevenson et al.	N/A	N/A
10099051	12/2017	Stevenson et al.	N/A	N/A
10124164	12/2017	Stevenson et al.	N/A	N/A
10249415	12/2018	Seitz et al.	N/A	N/A
10272252	12/2018	Seitz et al.	N/A	N/A
10272253	12/2018	Seitz et al.	N/A	N/A
10306848	12/2018	Chan et al.	N/A	N/A
10350421	12/2018	Stevenson et al.	N/A	N/A
10420949	12/2018	Seitz et al.	N/A	N/A
10449375	12/2018	Frustaci et al.	N/A	N/A
10499375	12/2018	Jung et al.	N/A	N/A

10500402	12/2018	Stevenson et al.	N/A	N/A
10559409	12/2019	Seitz et al.	N/A	N/A
10561837	12/2019	Stevenson et al.	N/A	N/A
10589107	12/2019	Seitz et al.	N/A	N/A
10596369	12/2019	Stevenson et al.	N/A	N/A
10722706	12/2019	Stevenson et al.	N/A	N/A
10828498	12/2019	Stevenson et al.	N/A	N/A
RE48348	12/2019	Stevenson	N/A	N/A
10857369	12/2019	Stevenson et al.	N/A	N/A
10874866	12/2019	Stevenson et al.	N/A	N/A
10905888	12/2020	Stevenson et al.	N/A	N/A
10912945	12/2020	Stevenson et al.	N/A	N/A
11013928	12/2020	Stevenson et al.	N/A	N/A
11071858	12/2020	Stevenson et al.	N/A	N/A
11147977	12/2020	Stevenson et al.	N/A	N/A
11185705	12/2020	Stevenson et al.	N/A	N/A
11198014	12/2020	Stevenson et al.	N/A	N/A
11211741	12/2020	Marzano et al.	N/A	N/A
11241581	12/2021	Stevenson et al.	N/A	N/A
11344734	12/2021	Stevenson et al.	N/A	N/A
11406817	12/2021	Stevenson et al.	N/A	N/A
11633612	12/2022	Frysz et al.	N/A	N/A
11648408	12/2022	Hussein	N/A	N/A
11764745	12/2022	Sahabi et al.	N/A	N/A
11980766	12/2023	Woods et al.	N/A	N/A
12246183	12/2024	Woods	N/A	H01R 13/521
12268889	12/2024	Woods	N/A	H01G 4/35
2003/0053294	12/2002	Stevenson	N/A	N/A
2009/0080140	12/2008	Iyer et al.	N/A	N/A
2011/0287644	12/2010	Kuwahara	439/110	H01M 50/522
2018/0216654	12/2017	Santos	N/A	F16B 19/02
2020/0116157	12/2019	Tanaka	N/A	F04D 29/626
2020/0200182	12/2019	Taki	N/A	F04D 17/16
2022/0115806	12/2021	Marzano et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3449973	12/2018	EP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 18/633,096, filed on Apr. 11, 2024, now U.S. Pat. No. 12,268,889, which is a continuation of U.S. application Ser. No. 18/631,709, filed on Apr. 10, 2024, now U.S. Pat. No. 12,246,183, which is a continuation-in-part of U.S. application Ser. No. 18/377,609, filed on Oct. 6, 2023, now U.S. Pat. No. . 11, 980, 766.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention generally relates to medical devices. More particularly, the present invention describes a self-centering polymeric or ceramic washer that is positioned between a hermetic feedthrough and a circuit board supporting a plurality of filter capacitors for a medical device, for example, an active implantable medical device.

2. Prior Art

(2) A polymeric washer for use in filtered feedthrough assemblies is typically made from a relatively lubricious polymeric material. However, lubricious (slippery) polymeric washers are particularly problematic with regards to centering their openings about a plurality of terminal pins of the feedthrough. Even if the polymeric washer has an adhesive coated on each of its opposed sides, because the washer must be applied and laminated to adjacent structures under significant pressure and high temperature, the washer can easily slide into a misaligned position with respect to the gold braze that hermetically seals each of the terminal pins extending through the via holes in the feedthrough insulator.

(3) Moreover, the problem of inadvertent misalignment as a washer is being applied and laminated to adjacent structures under pressure and high temperature is not unique to a polymeric washer. A ceramic washer, even one that has an adhesive coated on each of its opposed sides, can also be prone to sliding out of proper alignment as it is being laminated between the insulator of a feedthrough and an EMI filter capacitor or a circuit board supporting a plurality of EMI filter capacitors.

(4) Accordingly, a self-centering polymeric or ceramic washer that is configured to reside between a feedthrough and an EMI filter capacitor or a circuit board supporting a plurality of EMI filter capacitors and that has a number of shaped openings to prevent the washer from inadvertent lateral movement out of alignment on the feedthrough insulator is needed.

BACKGROUND OF THE INVENTION

(5) The present invention relates to a feedthrough assembly that is attachable to a circuit board to form an assembly for incorporation into an active implantable medical device (AIMD). The feedthrough assembly comprises a polymeric or ceramic self-centering washer that is disposed between a feedthrough and a circuit board that supports a plurality of filter capacitors, for example MLCC chip capacitors or X2Y attenuator capacitors.

(6) The self-centering washer has shaped openings through which terminal pins of the hermetic feedthrough extend. For example, with a feedthrough having first and second terminal pins, one embodiment of the self-centering washer has respective first and second shaped openings. The first shaped opening has a first inner arcuate or curved portion that contacts the first terminal pin and a first outer perimeter portion that is spaced from the first terminal pin. The first outer perimeter portion exposes a portion of the first braze hermetically sealing the first terminal pin to the insulator in an insulator first via hole. Similarly, the second shaped opening in the self-centering washer has a second inner arcuate or curved portion that contacts the second terminal pin and a second outer perimeter portion that is spaced from the second terminal pin. The second outer perimeter portion exposes a portion of the second braze hermetically sealing the second terminal pin to the insulator in an insulator second via hole.

(7) Then, in an imaginary configuration having the first and second shaped openings superimposed one on top of the other, the cumulative arcuate distance measured in degrees of the respective first and second inner arcuate portions about one of the first and second terminal pins, subtracting overlap, results in a gap between the superimposed first and second shaped openings that is less than a diameter of either of the first and second terminal pins. That way, the polymeric or ceramic self-centering washer of the present invention is prevented from any lateral movement with respect to a device side of the insulator.

(8) Preferably, the first and second inner arcuate portions of the first and second shaped openings each extend from about 90° to about 180° around the perimeter of the respective first and second terminal pins, and the first and second outer peripheral portions of the first and second shaped openings each extend from about 90° to about 180° around the perimeter of the respective first and second brazes. Further, the opposed ends of the inner arcuate portions meet opposed ends of the outer peripheral portions at rounded corners.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a wire formed diagram of a generic human body showing a number of medical devices **100A** to **100L** that can either be implanted in a patient's body or attached externally to the body.
- (2) FIG. 2 illustrates a side cutaway view of the cardiac pacemaker **100C** shown in FIG. 1.
- (3) FIG. 3 is a perspective view of a filter feedthrough **123** comprising a quad-polar EMI filter capacitor **122** mounted on a hermetically sealed feedthrough **120**.
- (4) FIG. 4 is a cross-sectional view taken along line 4-4 of FIG. 3.
- (5) FIG. 5 is a perspective view of a filter feedthrough **123** comprising an inline quad-polar EMI filter capacitor **158** mounted on a hermetically sealed feedthrough **120**.
- (6) FIG. 6 is a cross-sectional view showing a number of MLCC chip capacitors **214**, **216** and **218** mounted on a circuit board **200** supported on a hermetically sealed feedthrough **120**.
- (7) FIG. 6A is an enlarged view of the indicated section in FIG. 6.
- (8) FIG. 6B is a partial cross-sectional view of a ground terminal pin **228** welded or brazed **230** into an opening in the ferrule **124** and connected to the circuit board ground plates **202A**, **202B** by an electrically conductive material **234**.
- (9) FIG. 6C illustrates that the ground pin of FIG. 6B can be replaced by a ground via **236** that is spatially aligned over the hermetic seal to ferrule gold braze **132**.
- (10) FIG. 6D illustrates that the circuit board **200** of FIG. 6 may have an edge metallization **206** that is attached to a metal addition **250**, which is laser welded or gold brazed **252** to the ferrule **124**.
- (11) FIG. 6E illustrates that the circuit board **200** of FIG. 6 can have a ground edge metallization **206** that is connected to a gold pocket-pad **256**, which is formed into a recess in the ferrule **124**.
- (12) FIG. 6F is similar to FIG. 6E, except that the ground via hole **236** is spatially aligned over a gold pocket-pad **256** formed in a ferrule recess.
- (13) FIG. 6G illustrates that the circuit board **200** may have an edge ground metallization **206** that is conductively coupled to the oxide-resistant ground pin **228** shown in FIG. 6B.
- (14) FIG. 7 is a plan view of a hermetically sealed feedthrough **120** similar to the feedthroughs **120** shown in FIGS. 3 to 5 on which a prior art polymeric washer **152** is undesirably misaligned over gold brazes **138** for terminal pins **130A** to **130E**.
- (15) FIG. 8 is a cross-sectional view taken along line 8-8 of FIG. 7.
- (16) FIG. 8A is an enlarged view of the indicated section of FIG. 8.
- (17) FIG. 9 is a plan view of a novel self-centering polymeric or ceramic washer **170** according to the present invention.
- (18) FIG. 10 is an exploded view of a filtered feedthrough **123** that is similar to that shown in FIG. 5 but including the novel self-centering polymeric or ceramic washer **170** shown in FIG. 9 mounted on the hermetically sealed feedthrough **120**.
- (19) FIG. 11 is a plan view of a hermetically sealed feedthrough **120** on which the self-centering polymeric or ceramic washer **170** shown in FIGS. 9 and 10 is desirably located.
- (20) FIGS. 11A to 11D are sectional views of respective openings **172**, **174**, **176** and **180** of the polymeric or ceramic washer **170** shown in FIG. 11 self-centered about terminal pins **130A**, **130B**, **130C** and **130E**, respectively.

- (21) FIG. 11E shows openings **172** and **174** from FIGS. 11A and 11B superimposed one on top of the other.
- (22) FIG. 12 an isometric view of another embodiment of a novel self-centering polymeric or ceramic washer **300** that has shaped openings with two diametrically opposed inwardly extending tooth-like structures providing opposed inner arcuate portions according to the present invention.
- (23) FIG. 13 is a plan view of a hermetically sealed feedthrough **120** on which a novel self-centering polymeric or ceramic washer **320** having shaped openings with four inwardly extending tooth-like structures providing four inner arcuate portions according to the present invention.
- (24) FIG. 14 is a plan view of a hermetically sealed feedthrough **120** on which a self-centering polymeric or ceramic washer **350** is desirably located so that the exposed insulator-to-terminal pin gold brazes **138** are outside of high-voltage keep-out zones **360** and **362**.
- (25) FIG. 15 is a plan view of a hermetically sealed feedthrough **120** on which a self-centering polymeric or ceramic washer **350A** with terminal pins **130A** to **130D** received in baseball playing field-shaped openings for so that the exposed insulator to terminal pin gold brazes **138** are outside a narrower high-voltage keep-out zone **364** as compared to the embodiment shown in FIG. 15.
- (26) FIG. 16 is a perspective view of the MLCC chip capacitor **214** as a representative one of the chip capacitors mounted on the circuit board **200** shown in FIG. 6.
- (27) FIG. 16A is a cross-sectional view taken along line **16A-16A** of FIG. 16.
- (28) FIG. 16B is a cross-sectional view taken along line **16B-16B** of FIG. 16A.
- (29) FIG. 16C is a cross-sectional view taken along line **16C-16C** of FIG. 16A.
- (30) FIG. 17 is a perspective view of an X2Y attenuator **240** that is useful in lieu of the MLCC chip capacitors **214** to **218** mounted on the circuit board **200** shown in FIG. 6.
- (31) FIG. 18 is a sectional view taken along line **18-18** of FIG. 17.
- (32) FIG. 19A is an elevational view illustrating the novel self-centering washer **170** shown in FIGS. 6, 6A, 6B, 9 to 11 and 11A to 11E and the self-centering washers **300**, **320**, **350** and **350A** shown in respective FIGS. 12 to 15 comprising a polymeric or ceramic material.
- (33) FIG. 19B is an elevational view illustrating an embodiment of the self-centering polymeric washers **170**, **300**, **320**, **350** and **350A** shown in FIG. 19A containing insulating nanoparticles.
- (34) FIG. 19C is an elevational view illustrating the self-centering polymeric or ceramic washers **170**, **300**, **320**, **350** and **350A** shown in FIG. 19A having an adhesive **266A** disposed on only one surface of the washer.
- (35) FIG. 19D is an elevational view illustrating the self-centering polymeric or ceramic washers **170**, **300**, **320**, **350** and **350A** shown in FIG. 19A having either an adhesive **266A** or a thermoplastic material **266B** disposed on both surfaces of the washer.
- (36) FIG. 19E is an elevational view illustrating two of the self-centering polymeric or ceramic washers **170**, **300**, **320**, **350** and **350A** shown in FIG. 19A secured together by an intermediate adhesive **266A** or thermoplastic layer **266B** and being coated on their respective outer surfaces with an adhesive **266A** or thermoplastic material **266B**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (37) FIG. 1 is a wireframe diagram of a generic human body showing a number of active medical devices. An active medical device has at least one power source and at least one electronic circuit.
- (38) Numerical designation **100A** is a family of external and implantable hearing devices, which include the group of hearing aids, cochlear implants, piezoelectric sound bridge transducers and the like.
- (39) Numerical designation **100B** includes the family of neurostimulators and brain stimulators. For example, neurostimulators are used to stimulate the Vagus nerve to treat epilepsy, obesity, and depression. Brain stimulators are similar to a pacemaker-like device and include electrodes implanted deep into the brain for sensing the onset of a seizure and also providing electrical stimulation to brain tissue to prevent the seizure from actually occurring. The leads that come from a deep brain stimulator are often placed using real-time imaging. Frequently, such leads are placed

real-time using an MRI machine.

(40) Numerical designation **100C** shows a cardiac pacemaker, which is well-known in the art and may have either endocardial or epicardial leads. Implantable pacemakers may also be leadless (meaning without a lead or leads). The family of cardiac pacemakers **100C** includes cardiac resynchronization therapy devices (CRT-P pacemakers) and leadless pacemakers. CRT-P pacemakers are unique in that they pace both the right and left sides of the heart. The cardiac device family also includes any and all types of biologic monitoring and/or data recording devices and all types of implantable loop recorders (ILR) or other such monitors and recorders of biologic activity, for example, an ILR that records the electrical activity of the heart.

(41) Numerical designation **100D** includes the family of left ventricular assist devices (LVAD's) and artificial hearts.

(42) Numerical designation **100E** includes the family of drug pumps, which can be used for dispensing of insulin, chemotherapy drugs, pain medications, and the like. Insulin pumps are evolving from passive devices to active devices that have sensors and closed loop systems, which can, for example, monitor blood glucose levels in real time. Such active pump devices tend to be more sensitive to EMI than passive pumps, which have no sense circuitry or externally implanted leads.

(43) Numerical designation **100F** includes a variety of external or implantable bone growth stimulators for rapid healing of fractures.

(44) Numerical designation **100G** includes urinary incontinence devices.

(45) Numerical designation **100H** includes the family of pain relief spinal cord stimulators and anti-tremor stimulators. Numerical designation **100H** also includes the complete family of neurostimulators used to block pain signals.

(46) Numerical designation **100I** includes the families of implantable cardioverter defibrillators (ICD) and congestive heart failure (CHF) devices, including cardio-resynchronization therapy devices (CRT-D). A CRT-D, which is a special subcutaneous device for heart failure patients who are also at high risk for sudden cardiac death, can provide high-voltage defibrillation. The devices of numerical designation **100I** may have either endocardial or epicardial leads.

(47) Numerical designation **100J** illustrates an externally worn pack, such as, but not limited to, an external insulin pump, an external drug pump, an external neurostimulator, a Holter monitor with skin electrodes or a ventricular assist device power pack.

(48) FIG. 2 illustrates a side cutaway view of the cardiac pacemaker **100C** shown in FIG. 1. The pacemaker electronics are housed inside a hermetically sealed AIMD housing **102** (typically titanium), which provides an electrically conductive electromagnetic shield. The header **104** is supported on the housing **102** and is typically made from a thermosetting plastic, such as Tecothane®. The header **104** houses one or more connector assemblies that are generally in accordance with ISO Standards IS-1, IS-2, IS4 or DF4. The connector assemblies comprise terminal blocks (female connectors) that are labelled **106**, **106A**. Electrical connectors **108**, **108A** (male connectors) are located at the proximal end of implantable leads **110**, **110A**. The electrical connectors **108**, **108A** are designed to be inserted into and mate with the female terminal blocks **106**, **106A** in the header **104**. Distal and ring electrodes **112**, **112A** and **114**, **114A** are located at the distal end of the respective leads **110**, **110A**. The electrodes **112**, **112A** and **114**, **114A** are designed to contact body tissue, for example, the illustrated heart **116** to at least one of deliver electrical stimulation to the body tissue or sense biological signals from body tissue **116**.

(49) Further regarding FIG. 2, the AIMD housing **102** serves as a system ground **118** that provides the overall electromagnetic shield, and also functions as an otherwise dangerous EMI energy dissipating surface. The AIMD also has a hermetically sealed feedthrough **120** to which a quad-polar EMI filter capacitor **122** is mounted to thereby form a filtered feedthrough **123**. Since the conductive ferrule **124** of the feedthrough **120** is electrically connected to the AIMD housing **102**, the ferrule **124** of the feedthrough **120** is also part of the system ground **118**. Accordingly, as

illustrated, the system ground **118** illustrated in FIG. **2** includes the ferrule **124** and the device housing **102**.

(50) As shown in greater detail in FIGS. **3** and **4**, the hermetic feedthrough **120** comprises the ferrule **124** having an opening in which a ceramic insulator **126** resides. The ferrule **124** has an H-flange **124A** that serves to connect the feedthrough **120** to an opening in the device housing **102**. The H flange **124A** can comprise other shapes, as is known in the prior art. The insulator **126** has a number of via holes **128** through which terminal pins **130** extend. The insulator **126** also has an outer perimeter surface that is sized and shaped to reside inside the ferrule opening.

(51) A ring-shaped braze **132**, preferably a gold braze, hermetically seals the ferrule **124** to the outer perimeter surface of the insulator **126**. However, before the insulator **126** is brazed to the ferrule **124**, an adhesion layer **134** is directly applied to the outer surface of the insulator **126**, and a wetting layer **136** is applied on top of the outer adhesion layer **134**. The thusly prepared insulator **126** is then hermetically sealed to the ferrule **124** using the gold braze **132**.

(52) In a similar manner, an adhesion layer **134** is first directly applied to the inner surface of the insulator in the via holes, and the inner wetting layer **136** is applied on top of the inner adhesion layer **130**. Then, ring-shaped gold brazes **138** hermetically seal the exemplary number of four terminal pins **130** to the insulator **126** in the via holes **128**. In a preferred embodiment, the inner and outer adhesion layers **134** are titanium and the inner and outer wetting layers **136** are either molybdenum or niobium.

(53) Still referring to FIGS. **3** and **4**, the discoidal EMI filter capacitor **122** supported on the hermetic feedthrough **120** comprises a disc-shaped dielectric substrate **140** that supports a number of active electrode plates **142** that are interleaved in a partial overlapping arrangement with a number of ground electrode plates **144**. The dielectric substrate **140** has an outer sidewall and internal passageways through which the terminal pins **130** extend. An active termination material **146** in the dielectric substrate passageways contacts the active electrode plates **142** and a ground termination material **148** on the dielectric substrate outer sidewall contacts the ground electrode plates **144**. Then, an electrically conductive material **150A** filled into the capacitor passageways connects the active electrode plates **142**/active termination material **146** to the terminal pins **130**. Similarly, an electrically conductive material **150B** connects the ground electrode plates **144**/ground termination material **148** to the ferrule **124**. In a preferred embodiment, the electrical connection materials **150A**, **150B** are a polyimide, a solder, a thermal-setting conductive polymer such as a conductive epoxy, and the like.

(54) A polymeric washer **152** resides between the feedthrough **120**, particularly the insulator **126**, and the discoidal EMI filter capacitor **122**. The polymeric washer **152** is important because it helps prevent high voltage flashover, such as is possible in an implantable defibrillator application between terminal pins of opposite polarity and between an active terminal pin and the system ground including the conductive ferrule **124** of the feedthrough **120** which is electrically connected to the AIMD housing **102**.

(55) When the external capacitor metallization or ground termination material **148** of the discoidal filter capacitor **122** is electrically connected to the ferrule **124** of the hermetically sealed feedthrough **120**, and the ferrule **124** in turn is hermetically sealed (typically by welding, such as laser welding) to the conductive housing **102** of an AIMD, for example, a cardiac pacemaker **100C**, then the external capacitor ground termination material **148** is also electrically connected to the system ground **118**. This means that the external capacitor ground termination material **148**, the ferrule **124** and the AIMD housing **102** are all at the same ground potential and are all part of the overall AIMD equipotential surface (in other words, system ground **118**). Since the hermetically sealed enclosure of the AIMD blocks EMI from entering inside the device housing **102**, it is commonly known as an EMI shield or a Faraday cage.

(56) Additionally, the discoidal EMI filter capacitor **122** is designed to selectively redirect undesirable high-frequency energy before it enters the AIMD housing **102** to this equipotential

surface for diversion and/or energy dissipation. Ideally, the role of the feedthrough EMI filter capacitor **122** is designed to let low frequency biologic signals, such as therapeutic pacing pulses, freely pass without attenuation, while at the same time diverting dangerous high-frequency EMI energy to the AIMD housing **102** serving as an equipotential surface. The term “low frequency” refers to frequencies that range from a few hertz to kilohertz. Higher frequencies include about 1 MHz to about 200 MHz, which is in the range of the RF pulsed frequencies of 1.5 Tesla (64 MHz) and 3 Tesla (128 MHz) MRI scanners. The EMI filters of the present invention attenuate MRI RF frequencies very effectively.

(57) When undesirable EMI energy is diverted to the AIMD housing **102**, the undesirable EMI energy is dissipated harmlessly as a few milliwatts of heat energy (for an MRI scanner this can also be a few watts which is also harmless). In this manner, dangerous EMI energy is prevented from entering the AIMD housing **102** before the EMI energy can reach sensitive AIMD circuitry **154** mounted on the circuit board **156** (FIG. 2) housed inside the device housing **102**. Should EMI energy undesirably enter inside the device housing **102**, it could seriously disrupt the proper operation of the various AIMD circuits **154**. Such EMI disruption could inadvertently suspend therapy, which could be immediately life threatening.

(58) FIG. 5 illustrates an inline quad-polar EMI filter capacitor **158** mounted on the previously described hermetically sealed feedthrough **120** to thereby form a filtered feedthrough **160**. With the exception that it has a rectangular shape and not the disc-shape of the discoidal EMI filter capacitor **122** shown in FIGS. 3 and 4, EMI filter capacitor **158** is similar to the previously described discoidal EMI filter capacitor **122**. Electrical connection material **150B** connects the filter capacitor ground metallization **148** connected to ground electrode plates to the ferrule **124**, which is generally of titanium. The ground electrical connections are desirably very low impedance and very low resistance. In this design, the ground electrical connection material **150B** is connected to a gold bond pad **161**, which forms an oxide-resistant electrical connection to the ferrule **124**. Oxide-resistant EMI filter connections are described in U.S. Pat. No. 6,765,779, which is assigned to the assignee of the present invention and incorporated herein fully by reference. U.S. Pat. Nos. 9,427,596, 9,931,514, 10,350,421, 11,241,581, 11,344,734, and 11,633,612 also teach oxide-resistant low impedance electrical connections to a titanium ferrule and are also fully incorporated herein by reference. While not shown in the drawing of FIG. 5, a polymeric washer, similar to the washer **152** shown in FIG. 4, resides between the feedthrough **120**, particularly the insulator **126**, and the inline quad-polar EMI filter capacitor **158**.

(59) Turning now to FIG. 6, an EMI filter circuit board **200** is shown supported on the device side of the previously described hermetic feedthrough **120**. The filter circuit board **200** is supported on the device side of the insulator **126** and has at least one ground plate **202**, which can be an internal ground plate **202A** or an external ground plate **202B**. If external, the ground plate **202B** is positioned adjacent to the device side of the insulator **126**. It is within the scope of the present invention that the circuit board **200** can support more than one internal ground plate **202A** as well as the external ground plate **202B**. If desired, some or all of the external ground plate **202B** and the internal ground plate **202A** can be replaced by ground circuit traces. Various types of circuit boards and grounding techniques are more thoroughly described in U.S. Pat. No. 8,195,295, which is assigned to the assignee of the present invention and incorporated herein by reference.

(60) The circuit board internal and external ground plates **202A**, **202B** are grounded to the ferrule **112** in an oxide resistant manner. A metallization **206** contacted to the opposed edges of the circuit board **200** contacts the internal and external ground plates **202A**, **202B**. Then, an electrically conductive material **208**, for example, a polyimide, a solder, a thermal-setting conductive polymer such as a conductive epoxy, and the like, electrically connects the edge metallization **206** to the gold braze **132** hermetically sealing the insulator **126** to the ferrule **124**. In that manner, the circuit board ground plates **202A**, **202b** are connected to the system ground **118** comprising the ferrule **124** and the device housing **102**.

(61) The circuit board **200** also has a number, for example six, passageways through which terminal pins **130A** to **130F** extend. The terminal pins **130A** to **130F** are sufficiently long so that a proximal pin portion extends outwardly beyond the device side of the circuit board **200** and a distal pin portion extends outwardly beyond a body fluid side of the insulator **126**. The terminal pins **130A** to **130F** are secured in their respective passageways using a metal eyelet **210** and an electrically conductive material **212** that connects the eyelet to the terminal pin (FIG. **6A**). The electrically conductive material **212** can be a polyimide, a solder, a thermal-setting conductive polymer such as a conductive epoxy, and the like.

(62) FIG. **6** further shows that there are a number of MLCC chip capacitors **214**, **216** and **218** mounted on the device side of the circuit board **200**. The MLCC chip capacitors each have an active termination spaced from a ground termination. As particularly shown in FIGS. **16** and **16A** to **16C**, exemplary MLCC chip capacitor **214** comprises a dielectric substrate **220** that supports a plurality of active electrode plate **222** interleaved in a capacitive relationship with a plurality of ground electrode plates **224**. The active electrode plates **222** are connected to an active termination **214A** spaced from a ground termination **214B** that is connected to the ground electrode plates **224**. The other MLCC chip capacitors **216** and **218** shown in FIG. **6** have a similar structure.

(63) As shown in FIGS. **6** and **6A**, a circuit trace **226** connects from the active termination **214A** for MLCC chip capacitor **214** to a metal insert **210**. The metal insert **210** is connected to terminal pin **130B** by an electrically conductive material **212**, for example, an electrically conductive epoxy or solder.

(64) FIG. **6B** is a cross-sectional view illustrating an embodiment of the circuit board **200** having a ferrule ground pin **228** that is electrically and mechanically connected to the ferrule **124** by a gold braze or a laser weld **230**. Ideally, ground pin **228** is of an oxide-resistant material, such as, but not limited to, platinum. This makes for an essentially oxide-free electrical connection **234** to a circuit board via metallization **206A** connected to the internal and external ground plates **202A**, **202B**.

(65) Still referring to FIG. **6B**, one will see that the circuit board via hole metallization **206A** is spatially aligned over the gold braze or a laser weld **230**. Throughout the present invention, via holes are provided with some sort of a conductive or metallization layer on the inside diameters. It is understood by one skilled in the art that the inside diameter of circuit board via holes can be metal eyelets, plated, metallized, or the like. In each case, the conductive or metallization layer of the via hole is electrically connected to one or more internal or external circuit board ground plates or circuit traces. In the case of FIG. **6B**, the circuit board metallization **206A** makes electrical contact to the internal and external circuit board ground plates **202A**, **202B**. In the case where the ground pin **228** is of a lower cost material such as tantalum, niobium, or titanium, then the gold braze or laser weld **230** must be exposed such that electrical connection material **234** can contact termination **206A** and the gold braze or laser weld **230** thereby forming an oxide resistant low impedance connection.

(66) FIG. **6C** illustrates an alternative embodiment of how the circuit board ground plates **202A**, **202B** are grounded to the ferrule **124**. In this case, a circuit board ground via hole **236** is spatially aligned over the gold braze **138** that hermetically seals the insulator **126** to the ferrule **124**. By spatially aligning the ground via hole **236** over the gold braze **132**, an essentially oxide-free electrical connection directly to the hermetic seal gold braze is made using electrical connection material **238**. Electrical connection material **238** can be a solder, a thermal-setting conductive adhesive, and the like. FIG. **6D** shows that an oxide-free electrical connection to the circuit board ground plates **202A**, **202B** can also be achieved by an oxide-resistant metal addition **250**, which is either gold brazed, or laser welded **252** to the ferrule **124**. Typically, this metal addition **250** is of an oxide-resistant material, such as platinum. The metal addition **250** is connected to the circuit board edge metallization **206** by an electrical connection material **254**.

(67) FIG. **6E** illustrates another method of forming an oxide-resistant electrical connection from the circuit board ground plates **202A**, **202B** to the ferrule **124**. In this case, there is a gold pocket-pad

256, which acts like a swimming pool moat into which a gold braze, or equivalent material is formed. This forms an oxide-resistant electrical connection between the ferrule **124** and the electrical connection material **254** connected to the circuit board ground edge metallization **206** and, in turn, to the circuit board ground plates **202A**, **202B**.

(68) FIG. **6F** is very similar to FIG. **6E** in that an oxide-resistant gold bond pad **256** has been provided in a ferrule pocket. In this case, the circuit board ground via hole **236** has been spatially aligned over the gold pocket-pad **256**, such that an electrical connection **238** is made to the oxide-resistant noble gold filling the pocket. In this way, the circuit board ground plates **202A**, **202B** have a very low resistance connection to system ground **118**, which is the ferrule ground **124** connected to the device housing **102**. Gold pocket-pads are disclosed in U.S. Pat. No. 10,350,421, the contents of which are incorporated fully herein by this reference. Gold pocket-pads may comprise other oxide-resistant materials such as platinum. Noble metals, such as gold and platinum, are used as jewelry for this reason, as gold and platinum do not tarnish or oxidize over time. The pocket-pad **256** may comprise a number of other oxide-resistant materials, such as gold, gold alloys, rhodium, rhodium alloys, platinum, platinum alloys, platinum-iridium alloys, palladium, palladium alloys, nitinol, cobalt-chromium alloys, and combinations thereof.

(69) FIG. **6G** is similar to FIG. **6B** in that there is an oxide-resistant ground pin **228**, which is electrically and mechanically connected to the ferrule **124** by a laser weld or a gold braze **230** (gold brazes and laser welds burn through any oxide on the titanium ferrule forming a very stable and low resistance connection). In this case, instead of grounding through a via hole, as illustrated in FIG. **6B**, there is an electrical connection **234** from the circuit board ground edge metallization **206** directly to the oxide-resistant ground pin **228**. Accordingly, this provides a very low equivalent series resistance (low ESR) grounding path to system ground **118**. As previously defined, system ground **118** comprises the ferrule **124** connected to the device housing **102**. Importantly, the circuit board ground plates **202A**, **202B** provide a low impedance path for the EMI filters (an MLCC chip capacitor **214**, an X2Y attenuator **240** shown in FIGS. **16** and **16A** to **16C**, and combinations thereof) to divert dangerous EMI currents while at the same time shielding the insulator **126** from direct penetration of high frequency RE-radiated noise (EMI).

(70) Referring again to FIGS. **6** and **6B** to **6G**, the circuit board ground plates **202A**, **202B** can also be called circuit board ground shield plates. In other words, the ground plates **202A**, **202B** not only provide a low impedance filter circuit diversion pathway to system ground, but they also shield against EMI radiated energy from penetrating through the hermetic seal insulator **126** such that it could undesirably enter into the device housing **102**. Circuit board shield plates or ground plates act in an identical manner. Not shown is that these plates also absorb incident RF energy and that the capacitive action of the filter diverts the RF energy to the AIMD housing **102**, where it is harmlessly dissipated as a few milliwatts of heat energy. This diversion prevents the EMI from dangerously reaching the inside of the AIMD shielded housing **102**.

(71) Referring back to FIG. **6**, the ground termination **214B** for the MLCC chip capacitors is electrically connected to the ground plates **202A** and **202B** through a metal insert **232** that resides in a circuit board ground via. This metal insert **232** extends into the circuit board **200** to the depth of the ground plate, whether there is only an internal ground plate **202A**, only an external ground plate **202B** or both internal and external ground plates **202A**, **202B**.

(72) It bears mentioning that six terminal pins **130A**, **130B**, **130C**, **130D**, **130E** and **130F** are present in circuit board **200** depicted in FIG. **6**, but because of the cross-section, only three MLCC chip capacitors **214**, **216** and **218** are illustrated. This drawing is a cross-sectional view of the assembly so the other three MLCC chip capacitors connected to active terminal pins **130A**, **130C** and **130E** are not shown because they are mounted on the portion of the circuit board **200** that has been cut off by the cross-section. Nonetheless, MLCC chip capacitors **214**, **216** and **218** are integral to the illustrated assembly. MLCC chip capacitor EMI filters are more thoroughly described in U.S. Pat. No. 8,195,295, which is assigned to the assignee of the present invention and incorporated

herein by reference. Six active terminal pins **130A**, **130B**, **130C**, **130D**, **130E** and **130F** are just one illustration, as there can be any number of terminal pins depending on the particular active medical device design.

(73) In lieu of the MLCC chip capacitors **214** to **218** mounted on the circuit board **200** illustrated in FIG. **6**, it is within the scope of the present invention that all or some of those capacitors can be replaced by X2Y attenuators. An exemplary bipolar X2Y attenuator **240** is illustrated in FIGS. **17** and **18**. Bipolar means that the X2Y attenuator **240** can filter two leadwires at the same time. This is better understood by examining the internal active electrode plates **242A** and **242B** of FIG. **18**. The left-side active electrode plates **242A** are connected to the left side active capacitor metallization **244A**. The right-side active electrode plates **242B** are connected to the right-side active capacitor metallization **244B**. The active electrode plates **242A**, **242B** are interleaved with at least one ground electrode plate **246**.

(74) It is appreciated that any number of active and ground electrode plates can be interleaved to create a desired line-to-line or a line-to-ground amount of capacitance or filtering. For example, if a relatively large amount of line-to-line filtering (known as differential mode filtering) is desired, that is, filtering between active capacitor electrodes **242A** (Ca) and **242B** (Cb), then the interleaved ground electrode plate **246** is reduced in size (can be almost eliminated), which thus increases the effective capacitance area (ECA) between the two opposed active electrode plates **242A** (Ca) and **242B** (Cb). However, if mostly line-to-ground filtering is desired (known in the art as common mode filtering), filtering is from both active electrodes **242A** (Ca) and **242B** (Cb) to an enlarged ground electrode plate **246** (Gnd) metallization **244A**. The continuous ground metallization **248** is shown as a metallization band that is disposed all the way around the X2Y attenuator **240**. However, the ground metallization **248** can also be discontinuous as long as it electrically connects all of the ground plates to each other.

(75) Simultaneous filtering of two feedthrough leadwires is possible because the active electrode plates **242A** and **242B** of the X2Y attenuator **240** are designed such that the X2Y attenuator essentially comprises two different capacitors, one capacitor comprising the active electrode plates **242A** electrically connected to the active metallization **244A** and a second capacitor comprising the active electrode plates **242B** electrically connected to the active metallization **244B**. Each active electrode plate **242A**, **242B** sandwiched by the ground electrode plates **246** is connected to its respective active metallization **244A** and **244B**. This means that the active electrode plates **242A**, **242B** are electrically connectable to two of the leadwires **130A** to **130F** of the EMI filtering circuit board **200**. As such, one leadwire **130A** to **130F** is electrically connectable in pairs to the active electrode metallization **244A** and a second one of the leadwires **130A** to **130F** is electrically connectable to the active electrode metallization **244B**.

(76) Referring now to FIG. **7**, this drawing is a plan view of an exemplary polymeric washer **152** of the type that might reside between the feedthrough **120**, particularly the insulator **126**, and the discoidal EMI filter capacitor **122** in FIG. **4**, between the feedthrough **120** and the inline quad-polar EMI filter capacitor **158** in FIG. **5**, and between the feedthrough **120** and the circuit board **200** shown in FIG. **6**. This drawing shows that the prior art polymeric washer **152** has round-shaped openings **162** (which is typical) for receiving the respective feedthrough terminal pins **130A** to **130E**. However, the round-shaped openings **162** let the washer **152** move along the upper surface of the feedthrough insulator **126** and out of a centered alignment with respect to the terminal pins **130A** to **130E**.

(77) For example, using the x-y axis coordinate system shown in FIG. **7** for directional referencing, the polymeric washer **152** has undesirably moved upwardly along the y-axis so that a portion of the round-shaped openings **162** contact a lower side of the terminal pins **130A** to **130E** while an opposed portion of the washer openings **162** has slid past the insulator-to-ferrule braze **150B** so that an adjacent portion of the insulator **126** is uncovered. The polymeric washer **152** can move in a myriad of undesirable misaligned directions along the insulator **126** including downwardly along

the y-axis, to the left along the x-axis, or to the right along the x-axis. The problem is that when such sliding misalignment occurs, a portion of the feedthrough insulator **126** is left uncovered. That can lead to a high voltage flashover situation, such as is possible in an implantable defibrillator application between terminal pins of opposite polarity and between an active terminal pin and the system ground including the conductive ferrule **124** of the feedthrough **120** electrically connected to the AIMD housing **102**.

(78) The cross-sectional views of FIGS. **8** and **8A** further illustrate undesirable misalignment along the planar surface of the feedthrough insulator **126** with respect to a representative one of the terminal pins **130**. Misalignment of the polymeric washer **152** creates a gap **164** between the gold brazes **138** that hermetically seal the terminal pin **130** to the insulator **126** in the via hole **128** and the closest ground, which in this embodiment, is the ferrule gold braze **132**. The gap **164** can result in the highly undesirable effect of reducing the high-voltage keep-out zone **166** (FIG. **7**), which is also the flashover distance. The gap **164** can be particularly problematic when the feedthrough **120** is exposed to a high volts per mil (V/mil) stress. Partial delamination between the polymeric washer **152** and the feedthrough insulator **126** can create an air bubble between the positive and negative or active and ground poles, which can enable a voltage arc or avalanche. Moreover, the gap **164** within the keep-out zone **166** of the gold braze **132** can undesirably lead to a high-voltage discharge or a catastrophic avalanche discharge.

(79) Commensurate with the amount of misalignment that exists, the misaligned polymeric washer **152** will also undesirably cover a portion of the insulator-to-ferrule gold braze **132**. The covered portion of the gold braze **132** becomes unavailable for an oxide-resistant electrical connection.

(80) It is noted that the telemetry pin T in the embodiment shown in FIG. **7** is not associated with a filter capacitor and is therefore not covered by the washer **152**. The reason the polymeric washer **152** does not cover the telemetry pin T is because the telemetry pin cannot be filtered. RF telemetry requires a high frequency signal that will not communicate if it is filtered. Instead, the telemetry pin T enables high-frequency RF communication between an external AIMD programmer and the implanted AIMD. This means that the telemetry pin T must not have any electrical connection to an EMI filter capacitor **122** (FIGS. **3** and **4**), an EMI filter capacitor **158** (FIG. **5**) or an EMI filter circuit board **200** (FIG. **6**). Otherwise, an RF telemetry signal travelling down the telemetry pin T would be undesirably attenuated. Moreover, the telemetry pin T is typically located in the header **104** (FIG. **2**), which is outside the AIMD housing **102**, and it is relatively small in comparison to the implantable leads **110**, **110A** (FIG. **2**) or, the telemetry pin T is positioned in a separate area inside the AIMD. Accordingly, the telemetry pin T does not tend to pick up much EMI and, therefore, does not need EMI filtering.

(81) FIG. **9** is a plan view that illustrates one embodiment of a novel self-centering polymeric or ceramic washer **170** according to the present invention. The self-centering polymeric or ceramic washer **170** replaces the washer **152** shown in FIG. **7** and comprises four spaced-apart shaped openings **172** to **178** that extend through its thickness. As further shown in FIG. **10**, the openings **172** to **178** are sized and shaped to receive a corresponding one of the terminal pins **130A** to **130D** of the feedthrough **120** shown in FIG. **5**. Generally, the shaped openings **172** to **178** each comprise an opening inner arcuate portion that contacts the terminal pin and an opening outer perimeter portion that is spaced from the terminal pin and that exposes a portion of the braze hermetically sealing the terminal pin to the insulator in a via hole extending through the insulator **126**.

(82) It is noted that while four openings **172** to **178** are provided in the polymeric or ceramic washer **170** shown in FIGS. **9** and **10**, that is not intended to be a limitation of the present invention. A self-centering polymeric or ceramic washer according to the present invention will have as many spaced-apart openings as there are active terminal pins which will subsequently be connected to an EMI filter capacitor **122** (FIGS. **3** and **4**), an EMI filter capacitor **158** (FIG. **5**) or an EMI filter circuit board **200** (FIG. **6**) that are designed to selectively redirect undesirable high-frequency energy before it enters the AIMD housing **102**.

(83) FIG. 11 is a plan view of the self-centering polymeric or ceramic washer 170 shown in FIGS. 9 and 10 mounted on top of the feedthrough 120 shown in FIGS. 3 to 6. As previously described, the telemetry pin T is not filtered, so the polymeric or ceramic washer 170 does not extend to that pin (although it could as long as the filter capacitor or filter circuit board does not contact the telemetry pin T).

(84) FIGS. 11A, 11B, 11C and 11D are enlarged sectional plan views of the respective shaped openings 172, 174, 178 and 180 of the self-centering polymeric or ceramic washer 170. FIG. 11E is an enlarged sectional plan view illustrating the shaped openings 172 and 174 of respective FIGS. 11A and 11B superimposed one on top of the other. The reason for this will be described in detail hereinafter.

(85) Looking first at opening 172 in FIG. 11A, it has an outer perimeter portion 172A that extends for about 90° to about 180° around the circular perimeter of the gold braze 138 which resides underneath the polymeric or ceramic washer 170 at the terminal pin 130A. The outer perimeter portion 172A extends to opposed inwardly extending upper and lower edges 172B and 172C that in turn meet an inner arcuate portion 172D. The inner arcuate portion 172D faces the outer perimeter portion 172A and extends for about 90° to about 180° around the circular perimeter of the terminal pin 130A. The outer perimeter portion 172A meets the upper curved edge 172B at junction 172E and the lower curved edge 172C at junction 172F. Further, the upper curved edge 172B meets the inner arcuate portion 172D at junction 172G while the opposed lower curved edge 172C meets the inner arcuate portion 172D at junction 172H.

(86) FIG. 11A also shows an imaginary x and y-axis coordinate system in dashed lines that is centered on the terminal pin 130A. An imaginary vertical line 182 extending through junctions 172G and 172H intersects the x-axis to the left of the y-axis. With this orientation of the shaped opening 172, the inner arcuate portion 172D is centered along the x-axis at about 9 o'clock with respect to the clock face shown in FIG. 11 and contacts the perimeter of the terminal pin 130A through an arc of about 90° to about 180°. This positioning means that only with respect to opening 172, the polymeric or ceramic washer 170 is capable of lateral movement only in a leftwards direction along the x-axis. Movement to the right along the x-axis is blocked by the inner arcuate portion 172D contacting the terminal pin 130A while movement in an upwardly direction and a downwardly direction along the y-axis is blocked by those portions of the inner arcuate portion 172D that reside immediately adjacent to the junctions 172G and 172H, respectively.

(87) As particularly shown in FIG. 11B, opening 174 associated with terminal pin 130B has a shape that is similar to that of opening 172 about terminal pin 130A, except that its inner arcuate portion contacts the perimeter of the terminal pin 130B centered at about 12-o'clock with respect to the clock face shown in FIG. 11. Opening 174 has an outer perimeter portion 174A that extends from about 90° to about 180° about the circular perimeter of the gold braze 138 that resides underneath the polymeric or ceramic washer 170 at terminal pin 130B. The outer perimeter portion 174A extends to opposed inwardly extending right and left edges 174B and 174C that in turn extend to an inner arcuate portion 174D. The outer perimeter portion 174A meets the right curved edge 174B at junction 174E and the left curved edge 174C at junction 174F. Further, the right curved edge 174B meets the inner arcuate portion 174D at junction 174G while the opposed left curved edge 174C meets the inner arcuate portion 174D at junction 174H.

(88) FIG. 11B also shows an imaginary x-axis and y-axis coordinate system in dashed lines that is centered on the terminal pin 130B. An imaginary horizontal line 184 extending through junctions 174G and 174H intersects the y-axis above the y-axis. With this orientation of the shaped opening 174, the inner arcuate portion 174D is centered along the y-axis at about 12 o'clock with respect to the clock face shown in FIG. 11 and contacts the perimeter of the terminal pin 130B through an arc of about 90° to about 180°. This positioning means that only with respect to opening 174, the polymeric or ceramic washer 170 is capable of lateral movement in only an upwardly direction along the y-axis. Movement in a downwardly direction along the y-axis is blocked by the inner

arcuate portion **174D** contacting the terminal pin **130B** while movement in a rightwards direction and a leftwards direction along the x-axis is blocked by those portions of the inner arcuate portion **174D** that reside immediately adjacent to the junctions **174G** and **746H**, respectively.

(89) FIG. **11C** shows that the opening **176** associated with terminal pin **130C** has a shape that is similar to those of openings **172** and **174** about terminal pins **130A** and **130B**, except that its inner arcuate portion contacts the perimeter of the terminal pin **130C** centered at about 6-o'clock with respect to the clock face shown in FIG. **11**. Opening **176** has an outer perimeter portion **176A** that extends from about 90° to about 180° about the circular perimeter of the gold braze **138** that resides underneath the polymeric or ceramic washer **170** at terminal pin **130C**. The outer perimeter portion **176A** extends to opposed inwardly extending right and left edges **176B** and **176C** that in turn meet an inner arcuate portion **176D**. The outer perimeter portion **176A** meets the right curved edge **176B** at junction **176E** and the left curved edge **176C** at junction **176F**. Further, the right curved edge **176B** meets the inner arcuate portion **176D** at junction **176G** while the opposed left curved edge **176C** meets the inner arcuate portion **176D** at junction **176H**.

(90) FIG. **11C** also shows an imaginary x-axis and y-axis coordinate system in dashed lines that is centered on the terminal pin **130C**. An imaginary horizontal line **186** extending through junctions **176G** and **176H** intersects the y-axis below the x-axis. With this orientation of the shaped opening **176**, the inner arcuate portion **176D** is centered along the y-axis at about 6 o'clock with respect to the clock face shown in FIG. **11** and contacts the perimeter of the terminal pin **130C** through an arc of about 90° to about 180°. This positioning means that only with respect to opening **176**, the polymeric or ceramic washer **176** is capable of movement only in a downwardly direction along the y-axis. Movement in an upwardly direction along the y-axis is blocked by the inner arcuate portion **176D** contacting the terminal pin **130C** while movement in a rightwards direction and a leftwards direction along the x-axis is blocked by those portions of the inner arcuate portion **130D** that reside immediately adjacent to the junctions **176G** and **176H**, respectively.

(91) Opening **178** has a similar structure and orientation as previously described for opening **174**.

(92) FIG. **11D** shows that the opening **180** associated with terminal pin **130E** has a shape that is similar to those of openings **172**, **174**, **176** and **178** about terminal pins **130A**, **130B**, **130C** and **130D**, respectively, except that its inner arcuate portion contacts the perimeter of the terminal pin **130E** centered at about 3 o'clock with respect to the clock face shown in FIG. **11**. Opening **180** has an outer perimeter portion **180A** that extends from about 90° to about 180° about the circular perimeter of the gold braze **138** that resides underneath the polymeric or ceramic washer **170** at terminal pin **130E**. The outer perimeter portion **180A** extends to opposed inwardly extending right and left edges **180B** and **180C** that in turn extend to an inner arcuate portion **180D**. The outer perimeter portion **180A** meets the right curved edge **180B** at junction **180E** and the left curved edge **180C** at junction **180F**. Further, the right curved edge **180B** meets the inner arcuate portion **180D** at junction **180G** while the opposed left curved edge **180C** meets the inner arcuate portion **180D** at junction **180H**.

(93) FIG. **11D** also shows an imaginary x-axis and y-axis coordinate system in dashed lines that is centered on the terminal pin **130E**. An imaginary vertical line **188** extending through junctions **180G** and **180H** intersects the x-axis to the right of the y-axis. With this orientation of the shaped opening **180**, the inner arcuate portion **180D** is centered along the x-axis at about 3 o'clock with respect to the clock face shown in FIG. **11** and contacts the perimeter of the terminal pin **130E** through an arc of about 90° to about 180°. This positioning means that only with respect to opening **180**, the polymeric or ceramic washer **170** is capable of movement only in a rightwards direction along the x-axis. Movement in a leftwards direction along the x-axis is blocked by the inner arcuate portion **180D** contacting the terminal pin **130E** while movement in an upwards direction and a downwards direction along the y-axis is blocked by those portions of the inner arcuate portion **180D** that reside immediately adjacent to the junctions **180G** and **180H**, respectively.

(94) That way, with respect to an imaginary configuration where at least two of the shaped

openings **172**, **174** or **178**, **176** and **180** are superimposed one on top of the other and with respect to their orientations in the polymeric or ceramic washer **170**, if the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** of the at least two shaped openings about a terminal pin, subtracting overlap, results in a gap between the combined inner arcuate portions of the superimposed openings that is less than the diameter of any one of the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**), then the novel self-centering polymeric or ceramic washer **170** illustrated in FIGS. **9** to **11** and **11A** to **11D** is prevented from any lateral movement in a rightwards, leftwards, upwards, or downwards direction with respect to the feedthrough **120** shown in FIGS. **4** to **6**.

(95) For example, FIG. **11E** shows openings **172** and **174** superimposed on top of the other. The inner arcuate portion **172D** of opening **172** extends from junction **172G** to **172H**. The inner arcuate portion **174D** of opening **174** extends from junction **174G** to **176H**. As the drawings shows, the remaining portion **190** of one of the terminal pins **130A**, **130B** that is not contacted by one of the inner arcuate portions **172D** and **174D** extends from junction **172H** of opening **172** to junction **174G** of opening **174**. A straight line intersecting those junctions **172H**, **174G** results in a gap that is less than the diameter of the terminal pins **130A** and **130B**. This means that the polymeric or ceramic washer **170** is prevented from any lateral movement with respect to the device side of the insulator **120**. The same applies to terminal pins **130C** to **130F** (FIG. **6**).

(96) If, on the other hand, with respect to an imaginary configuration with at least two of the shaped openings **172**, **174** or **178**, **176** and **180** being superimposed one on top of the other so that the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** about a terminal pin **130**, subtracting overlap, results in a gap between the superimposed openings that is greater than the diameter of at least one of the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**), then a third one of the shaped openings **172**, **174** or **178**, **176** and **180** is needed. That is with respect to an imaginary configuration where at least three of the shaped openings **172**, **174** or **178**, **176** and **180** are superimposed one on top of the other and with respect to their orientations in the polymeric or ceramic washer **170** so that the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** about a terminal pin, subtracting overlap, results in a gap between the superimposed openings that is less than the diameter of any one the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**), then the novel self-centering polymeric or ceramic washer **170** illustrated in FIGS. **11**, **11A**, **11B**, **11C** and **11D** is prevented from any lateral movement in a rightwards, leftwards, upwards, or downwards direction with respect to the feedthrough **120**.

(97) Moreover, with respect to an imaginary configuration where there is no configuration with at least three of the shaped openings **172**, **174** or **178**, **176** and **180** being superimposed one on top of the other and with respect to their orientations in the washer **170** so that the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** about a terminal pin, subtracting overlap, results in a gap between the superimposed openings that is greater than the diameter of at least one of the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**), then a fourth one of the shaped openings **172**, **174** or **178**, **176** and **180** is needed. That is with respect to an imaginary configuration where at least four of the shaped openings **172**, **174** or **178**, **176** and **180** are superimposed one on top of the other and with respect to their orientations in the polymeric or ceramic washer **170** so that the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** about a terminal pin **130**, subtracting overlap, results in a gap between the superimposed openings that is less than the diameter of any one the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**), then the self-centering polymeric or ceramic washer **170** illustrated in FIGS. **11**, **11A**, **11B**, **11C** and **11D** is prevented from any lateral movement in a rightwards, leftwards,

upwards, or downwards direction with respect to the feedthrough **120**.

(98) Thus, it is within the scope of the present invention that in order to prevent lateral movement of the self-centering polymeric or ceramic washer **170** in a rightwards, leftwards, upwards, or downwards direction with respect to the planar faces of the insulator and ferrule device sides of the feedthrough **120** shown in FIGS. **3** to **6**, at least two of the shaped openings **172**, **174** or **178**, **176** and **180** are required. That is with respect to the orientations of the shaped openings **172**, **174** or **178**, **176** and **180** in the polymeric or ceramic washer **170** being superimposed one on top of the other in an imaginary configuration so that the combined or cumulative arcuate distance measured in degrees of the respective inner arcuate portions **172D**, **174D** or **178D**, **176D** and **180D** about a terminal pin **130**, subtracting overlap, results in a gap between the superimposed openings that is less than the diameter of at least one of the terminal pins **130A** to **130F** (assuming equal diameters for the terminal pins **130A** to **130F**). That way, the novel polymeric or ceramic washer **170** of the present invention resolves the washer mis-location or misalignment issues of the prior art polymeric washer **152** illustrated in FIG. **7**.

(99) Referring once again to FIG. **11**, the outer perimeter portions **172A**, **174A**, **176A**, **178A** and **180A** of the shaped openings **172**, **174**, **176**, **178** and **180** for the polymeric or ceramic washer **170** selectively expose the terminal pin-to-insulator gold brazes **138** at the respective terminal pins **130A** to **130F**. The exposed portions of the gold braze **138** allow for robust and reliable oxide-resistant electrical connections to an EMI filter capacitor **122** (FIGS. **3** and **4**), an EMI filter capacitor **158** (FIG. **5**) or an EMI filter circuit board **200** (FIG. **6**). Even though the gold brazes **138**, which hermetically seal the terminal pins **130A** to **130F** to the feedthrough insulator **126** are partially covered by the respective polymeric or ceramic washer openings **172**, **174**, **176**, **178** and **180**, there is enough exposed gold for a low resistance, low impedance electrical connection.

(100) In order for there to be a sufficient high-voltage stand-off distance, a keep-out zone **192** as the distance between the inner edge of the gold braze **132** sealing the insulator **126** to the ferrule **124** and the closest edge of one of the adjacent openings **172**, **174**, **176**, **178** and **180** is illustrated in FIG. **11**. For example, the polymeric or ceramic washer-shaped opening orientation of the five pole in-line terminal pins **130A** to **130E** of FIG. **11** can be a challenge for providing sufficient stand-off distance. In high-voltage AIMDs, such as an ICD, or when low voltage AIMDs are exposed to defibrillation paddles or an automatic external defibrillator (AED), the high-voltage keep-out zone **192** becomes very important. It is a well-known principle in high-voltage engineering that any sharp point tends to be an equipotential field-line stress concentrator. In fact, a needlepoint-shaped corner is actually one of the worst cases for high-voltage breakdown of air or insulation between conductors. In that respect, the sharper the corner, the greater the threat of dielectric breakdown.

(101) For example, with reference to terminal pins **130A**, **130C** and **130E** shown in FIG. **11**, the junction **172E** where the outer perimeter portion **172A** meets the upper curved edge **172B** of shaped opening **172** is shown having a relatively sharp point that is oriented at about 12 o'clock and pointed directly into the high-voltage keep-out zone **192**. Similarly, the outer perimeter portion **180A** of shaped opening **180** meets the left curved edge **180C** at junction **180F**, which is shown as a relatively sharp junction. While these junctions **172E** and **180F** are shown as being relatively sharp only for the sake of illustration, they illustrate that a relatively high volts per mil (V/mil) stress may occur where they exist. On the other hand, the shaped opening **176** about terminal pin **130C** has its outer perimeter portion **176A** having a relatively large radial contour. The large radial contour of the outer perimeter portion **176A** faces the gold braze **132**, which is also at the edge of the high-voltage keep-out zone **192**. Comparing the sharp corners **172E**, **180F** of the shaped openings **172**, **180** about the respective terminal pins **130A** and **130E** with respect to the large radial contour **176A** of the shaped opening **176** about terminal pin **130C**, it is apparent that the latter design reduces the V/mil stress and is therefore unlikely to exhibit high-voltage breakdown than the sharp corners **172E**, **180F** at terminal pins **130A**, **130E**. Dielectric breakdown can lead to catastrophic high-

voltage avalanche. That is the case even though the exposed gold braze portions **138** of the terminal pins **130A** and **130E** have a keep-out zone **192** with respect to the insulator-to-ferrule braze **132** of the same distance as that for terminal pin **130C**.

(102) FIG. **12** is an isometric view of another embodiment of a novel self-centering polymeric or ceramic washer **300** according to the present invention. The self-centering polymeric or ceramic washer **300** has uniquely shaped openings **302** to **312** that each have two diametrically opposed tooth-like structures or inwardly extending projections that provide opposed inner arcuate portions. In particular, opening **302** has opposed tooth-like projections **302A** and **302B**, opening **304** has tooth-like projections **304A**, **304B**, opening **306** has tooth-like projections **306A**, **306B**, opening **308** has tooth-like projections **308A**, **308B**, opening **310** has tooth-like projections **310A**, **310B**, and opening **312** has tooth-like projections **312A**, **312B**. The openings are oriented so that an imaginary line **314** centered between the tooth-like projections is aligned at 12 o'clock to 6 o'clock for openings **302**, **306** and **310**, and an imaginary line **316** that is centered between the tooth-like projections for openings **304**, **308** and **312** is aligned at 3 o'clock to 9 o'clock. The openings being aligned from 12 o'clock to 6 o'clock and from 3 o'clock to 9 o'clock are in an alternating sequence.

(103) When the polymeric or ceramic washer **300** is seated on top of a hermetically sealed feedthrough, for example the feedthrough **120** shown in FIGS. **3** to **6**, it is within the scope of the present invention that the inner arcuate portions of the tooth-like projections of each of the openings **302** to **312** are configured to contact a terminal pin. Then, if both of the spaced apart gaps between the opposed inner arcuate portions contacting a terminal pin are less than the diameter of the terminal pin, the washer **300** is prevented from any lateral movement with respect to the device side of the insulator **126** for the feedthrough **120**.

(104) FIG. **13** illustrates another embodiment of a self-centering polymeric or ceramic washer **320** according to the present invention. The polymeric or ceramic washer **320** is positioned on the insulator **126** for a hermetically sealed feedthrough **120** (FIGS. **3** to **6**). The self-centering polymeric or ceramic washer **320** is similar to the self-centering washer **300** shown in FIG. **12**, except that each of the shaped openings **322**, **324**, **326**, **328**, **330** and **332** comprises four internal tooth-like structures (instead of two) that contact respective feedthrough terminal pins **130A** to **130F**. The four tooth-like structures provide four inner arcuate portions that fixedly align the polymeric or ceramic washer **320** about the terminal pins **130A** to **130F**. In that respect, the self-centering washer **320** does not require a unique orientation when mounted on the feedthrough **120**. The four tooth-like structures for each opening, being symmetric, prevent inadvertent lateral misalignment of the washer **320** with respect to the device side of the insulator **126** for the feedthrough **120**.

(105) FIG. **14** is a plan view of another embodiment of a self-centering polymeric or ceramic washer **350** having a number of spaced-apart novel baseball playing field-shaped openings **352**, **354**, **356** and **358**. The washer **350** is positioned on top of the insulator **126** for the hermetically sealed feedthrough **120**. In this embodiment, each baseball playing field-shaped opening **352**, **354**, **356** and **358** is oriented so that none of the exposed terminal pin-to-insulator gold brazes **138** encroach into the increased distance keep-out zone **360** (i.e., none of these gold brazes **138** extend into the high-voltage keep-out zone **360**). In particular, the open portion of the gold braze adjacent to terminal pin **130A** is generally oriented at 6 o'clock, the open portion of the gold braze adjacent to terminal pin **130B** is oriented at 4:30 o'clock, the open portion of the gold braze adjacent to terminal pin **130C** is oriented at 7:30 o'clock, and the open portion of the gold braze adjacent to terminal pin **130D** is oriented at 6 o'clock.

(106) The increased keep-out zone **360** distance extends ideally to the external surface of each terminal pin **130A** to **130D** and to the insulator-to-ferrule gold braze **132**, thereby providing optimal high-voltage stand-off distance therebetween. Optimal high-voltage stand-off distance greatly improves insulative reliability by providing increased safety margins against high-voltage insults. Each shaped opening **352**, **354**, **356** and **358** of the self-centering polymeric or ceramic

washer **350** has a baseball playing field-shaped opening that has a minor arc **352A**, **354A**, **356A** and **358A** abutting its corresponding terminal pin **130A** to **130D**. At least two shaped openings must be selectively oriented to prevent the self-centering polymeric or ceramic washer **350** from undesirably moving in any lateral direction along the planar faces of the insulator **126** of the feedthrough **120**. As previously disclosed, EMI filter capacitor **122** or EMI filter circuit board **200** electrical connection to the exposed gold brazes **138** is important in order to provide oxide-resistant low impedance, low resistance active terminal pin electrical connections.

(107) Referring again to FIG. **14**, on the left side, another high-voltage keep-out zone **362** is illustrated. This keep-out zone **362** extends to the rounded corner **358B** of opening **358** at the exposed gold braze **138** and to the insulator-to-ferrule gold braze **132**. As disclosed earlier, an exposed gold braze **138** at rounded corner **358B** is preferred compared to the exemplary sharp corners **352B**, **352C**, **352D** and **352E** of opening **352** adjacent to terminal pin **130A**.

(108) FIG. **15** illustrates another embodiment of the self-centering polymeric or ceramic washer **350A** shown in FIG. **14**, except that the open portion of the gold braze adjacent to terminal pin **130B** is oriented at 3 o'clock, and the open portion of the gold braze adjacent to terminal pin **130D** is oriented at 6 o'clock. As can be seen, the clocking-wise extent illustrated in this embodiment prevents inadvertent movement of the polymeric or ceramic washer **350A** in any lateral direction along the planar faces of the insulator **126** and ferrule device sides of the feedthrough **120**.

(109) The clock-wise orientations shown in FIG. **15**, however, position the exposed gold brazes **138** at a narrower keep-out zone **364** in comparison to the wider keep-out zone **360** illustrated in FIG. **14**. The keep-out zone **364** shown in FIG. **15**, while reduced, is not reduced as much as, for example, the keep-out zone **192** shown in the polymeric or ceramic washer **170** illustrated in FIG. **11** (distance **192** is less than HV stand-off distance **364**). Corners **354B**, **354C**, **354D**, **354E**, **356B**, **356C**, **356D**, **356E**, **358B**, **358C**, **358D** and **358E** are radiused which helps reduce volts/mil stress. That is in contrast to the sharp-edged corners **352B**, **352C**, **352D** and **352E** for shaped opening **352**, which are undesirable.

(110) Referring again to FIG. **15**, one also has to be concerned about the keep-out zone **362**, which has been previously discussed. In this case, keep-out zone **362** is adjacent to rounded corner **358C** of shaped opening at terminal pin **130D**. The rounded corner **358C** in comparison to the sharp corners **352B**, **352C**, **352D** and **352E** is preferred. That is because the rounded corners have reduced volts/mil stress.

(111) Referring now to FIG. **19A**, this drawing shows an elevational view illustrating the novel self-centering polymeric or ceramic washer **170** shown in FIGS. **6**, **6A**, **6B**, **9** to **11** and **11A** to **11E** and the self-centering polymeric or ceramic washers **300**, **320**, **350** and **350A** shown in respective FIGS. **12** to **15** comprising a homogeneous thermoplastic or ceramic. If polymeric, the washers **300**, **320**, **350** and **350A** may be made using conventional punch and cut methods from a flexible or semi-rigid film, tape, or equivalent insulating material (low plasticity, high elasticity), or they may be made from a nearly rigid polymeric or ceramic material.

(112) Suitable polymeric materials for the washers **170**, **300**, **320**, **350** and **350A** include polyimide (for example, Kapton®), aromatic, semi-aromatic, aliphatic, partially fluorinated aliphatic, acrilates, epoxies, elastomers, phenolics, polyimides, polyolephins, and fluoropolymers. Other suitable polymeric materials include silicone, polyurethane, polyester, polyethylene, polypropylene, polyamide (including synthetic polyamide, also known as nylon), acrylic, polyacrylates, and combinations thereof. Additional electrically insulative materials that are suitable for the washers **170**, **300**, **320**, **350** and **350A** include flowable perfluoroalkoxy (PFA), fluorinated ethylene-propylene (FEP), polyetheretherketone (PEEK), polyamide imide (PAI), polyphenyl sulfone (PPSU), polyetherimide (PEI), polymethyl methacrylate (PMMA), acrylonitrile butadiene styrene (ABS), polycarbonate (PC), polyoxymethylene (POM), polystyrene (PS), thermoplastic elastomer (TPE), polyethylene terephthalate (PET), ethylene-vinyl copolymers including ethylene-vinyl acetate (EVA) or polyethylene-vinyl acetate (PEVA), polytetrafluoroethylene, ethylene

tetrafluoroethylene, polyetherimide, acetal, polyacetal, polyformaldehyde, phenolic, polysulfide, and combinations thereof. Kapton® is a registered trademark of DuPont for polyimide films used as substrates for flexible printed circuits.

(113) Suitable ceramic materials for the washers **170, 300, 320, 350** and **350A** are selected from Al.sub.2O.sub.3, Bao, Cao, CeO.sub.2, MgO, ZrO.sub.2, SiO.sub.2, TiO.sub.2, Al.sub.2SiO.sub.5, BaTiO.sub.3, SrTiO.sub.2, ZnO, Si.sub.3N.sub.4, AlN, BN and combinations thereof. Various stabilized or partially stabilized zirconia may also be used including zirconia toughened alumina (ZTA) and alumina toughened zirconia (ATZ), yttrium stabilized zirconia (YSZ), yttrium-toughened zirconia (YTZP), and combinations thereof. Some nitrides may also be used, such as, aluminum nitride (AlN), silicon nitride (Si.sub.3N.sub.4), boron nitride (BN), carbon nitride (CN) and combinations thereof. Alumina (Al.sub.2O.sub.3) is a preferred material for the washers **170, 300, 320, 350** and **350A**.

(114) FIG. **19B** is an elevational view illustrating an embodiment of the self-centering polymeric washers **170, 300, 320, 350** and **350A** shown in FIG. **19A** but containing insulating nanoparticles **264**. The insulating nanoparticles **264** may be polymeric, ceramic or combinations thereof. Polymeric nanoparticle **264** may be selected from any of the insulating materials listed above for use as the polymeric washers **170, 300, 320, 350** and **350A**.

(115) Ceramic nanoparticles **264** may be selected from alumina, baria, calcia, ceria, magnesia, silica, strontia, titania, zirconia ceramic families and combinations thereof. Non-limiting examples of suitable nanoscale metal oxides that can also be used include Al.sub.2O.sub.3, Bao, Cao, CeO.sub.2, MgO, ZrO.sub.2, SiO.sub.2, TiO.sub.2, Al.sub.2SiO.sub.5, BaTiO.sub.3, SrTiO.sub.2, ZnO, Si.sub.3N.sub.4, AlN, BN and combinations thereof. Various stabilized or partially stabilized zirconia may also be used including zirconia toughened alumina (ZTA) and alumina toughened zirconia (ATZ), yttrium stabilized zirconia (YSZ), yttrium-toughened zirconia (YTZP), and combinations thereof. Some nitrides may also be used, such as, aluminum nitride (AlN), silicon nitride (Si.sub.3N.sub.4), boron nitride (BN), carbon nitride (CN) and combinations thereof.

(116) The nanoparticles **264** may be configured as particulates, short fibers, long fibers, spheres, flakes, submicron fibers, which are isotropically dispersed within the base polymeric material. It is understood that the nanoparticles **264** may be of the same material, or, alternatively, they may be a combination of one or more of the suitable materials listed above. For example, a combination of various nanoparticles **264** may be used. Insulating nanoparticle **264** material selection is defined by the specific electrical performance needs of an application and the physical, chemical, and electrical properties of the nanoparticle material. The above nanoparticles and particle configurations are applicable to all polymeric washer embodiments disclosed herein.

(117) The insulative nanoparticles **264** are represented by small uniform dots in FIG. **19B** that are evenly distributed throughout the various polymeric washers **170, 300, 320, 350** and **350A**. While small dots are used to indicate the nanoparticles **264**, in reality, the nanoparticles **264** are so small that they are not visible to the naked eye (requires microscopy techniques). The small dots are intended to distinguish the embodiments having nanoparticles according to the present invention from those embodiments that do not.

(118) Regarding particle size, the nanoparticles **264** range from greater than 0 nanometers to about 40,000 nanometers. As such, the nanoparticles **264** can range from greater than 0 nanometers to about 100 nanometers, which is the common industry range for the term “nano”. The particle size range of the nanoparticles **264**, however, also includes fine and micron-sized particles ranging from greater than about 100 nanometers (0.1 microns) to about 40,000 nanometers (40 microns).

(119) If the self-centering washers **170, 300, 320, 350** and **350A** are made of ceramic, there is any need for there to be added nanoparticles. Ceramic materials, for example, the preferred ceramic alumina, are inherently highly insulative.

(120) As further shown in FIGS. **19C** and **19D**, the polymeric or ceramic washers **170, 300, 320, 350** and **350A** may be a single or double-sided adhesive-backed **266A** washer or a thermoplastic-

backed **266B** washer. Suitable adhesives **266A** include reactive, non-reactive, pressure-sensitive, a synthetic or natural adhesives selected from acrylics, acrylates, acrylonitriles, cyanoacrylates, polyurethanes, polyvinyl acetates, polyvinyl alcohols, polyvinyl chlorides, polyesters, phenol-formaldehydes, polyamides, polyethylenes, polypropylenes, polysulfides, polyvinyl pyrrolidones, epoxies, styrene-butadienes, styrene acrylic copolymeric, silicones, silyl modified polymerics, among others. The adhesive **266A** may also include biocompatible, biostable, and non-toxic materials in the family of polyimides, polyamides, polyethylene terephthalates (PET), polydimethylsiloxane (PDMS), polytetrafluoroethylene (PTFE), ethylene tetrafluoroethylene (ETFE), parylenes, polyether block amides (PEBAX), polyetheretherketones (PEEK), polystyrenes, polysulfones, polypropylenes, polycarbonates, polyvinyl chlorides (PVC), polyxylylene polymerics, silicones, including medical-grade adhesives and epoxies.

(121) Suitable thermoplastic materials include a thermoplastic polyurethane, an aliphatic polyester-based thermoplastic polyurethane, an aliphatic and aromatic polycarbonate-based thermoplastic polyurethane, an aromatic polyether-based thermoplastic polyurethane. The term “thermoplastic” refers to any plastic or polymeric material that becomes pliable or moldable at a certain elevated temperature and solidifies on cooling. Thermoplastic or polymeric materials may be transformed into thermosetting plastic and polymeric materials by free radical cross-linking using techniques, such as, but not limited to, redox initiators or high energy radiation.

(122) Now looking at FIG. **19E**, a laminated configuration of two polymeric or ceramic washers is shown. The washers are separated by an adhesive layer **266A** or a thermoplastic layer **266B**. There is also an adhesive layer **266A** or a thermoplastic layer **266B** on the outer surfaces of the laminated washer. Importantly, in a laminated configuration, there must not be any separation, delamination, disconnection, disjunction, holes, openings, breaks, cavities, spaces, gas bubbles, or interruptions between the spaced-apart washers that could be a gas gap between the feedthrough insulator **126** and the EMI filter capacitors (FIGS. **3** to **5**) or the circuit board **200** (FIG. **6**). Proper bonding of the polymeric washers is therefore very important. Adjunct adhesives, epoxies, or polyimides (not shown) can facilitate and improve such bonding. Adding the previously described nanoparticles **264** to the polymeric washers can significantly increase the dielectric breakdown strength of the washer to greater than about 5 to about 10 thousand volts per mil thickness.

(123) In a preferred embodiment, the polymeric washers **170**, **300**, **320**, **350** and **350A** of the present invention have a CTE of approximately $8 \times 10^{-6}/^{\circ}\text{C.}$ at temperatures between 50°C. to 150°C. The CTE of the polymeric washers ranges from about $\geq 6 \times 10^{-6}/^{\circ}\text{C.}$ to $\leq 12 \times 10^{-6}/^{\circ}\text{C.}$ An embodiment of the present invention comprises a polyimide washer having a high temperature stability (glass transition temperature of approximately 300°C.) and a ring molecule to absorb mechanical and thermal stresses. The ability to withstand mechanical and thermal stresses is important for both thermal shock and piezoelectric stress induced purposes in ceramic filter capacitors during fast risetime ICD pulses.

(124) Thus, the present invention describes several embodiments of self-centering polymeric or ceramic washers that are configured to be positioned between the insulator of a feedthrough and either an EMI filter capacitor or a circuit board supporting a plurality of EMI filter capacitors. The washers are received over the terminal pins of the feedthrough and have self-centering structures that prevent them from inadvertent lateral movement on the insulator with respect to the terminal pins. One particularly preferred embodiment of a self-centering polymeric or ceramic washer according to the present invention has at least a first shaped opening and a second shaped opening through which first and second terminal pins extend. The first and second shaped openings each have an inner arcuate or curved portion that contacts the respective terminal pin and an outer perimeter portion that exposes the braze sealing the terminal pin to the feedthrough insulator. In an imaginary configuration with the first and second shaped openings superimposed one on top of the other, the cumulative arcuate distance measured in degrees of the inner arcuate portions about one of the terminal pins, subtracting overlap, results in a gap between the superimposed shaped

openings that is less than a diameter of either of the first and second terminal pins so that the self-centering polymeric or ceramic washer is prevented from lateral movement along the planar face of the feedthrough insulator.

(125) It is appreciated that various modifications to the inventive concepts described herein may be apparent to those skilled in the art without departing from the spirit and scope of the present invention as defined by the hereinafter appended claims.

Claims

1. A ceramic washer, comprising: a washer first opening comprising a washer first opening inner arcuate portion that is spaced from a washer first opening outer perimeter portion, wherein the washer first opening inner arcuate portion extends to opposed inner arcuate portion first and third ends and the washer first opening outer perimeter portion extends to opposed outer perimeter portion second and fourth ends, and wherein a first curved edge of the washer first opening meets the first and second ends of the respective inner arcuate and outer perimeter portions at first and second junctions, and a second curved edge of the washer first opening meets the third and fourth ends of the respective inner arcuate and outer perimeter portions at third and fourth junctions; and b) a washer second opening comprising a washer second opening inner arcuate portion that is spaced from a washer second opening outer perimeter portion, wherein the washer second opening inner arcuate portion extends to opposed inner arcuate portion fifth and seventh ends and the washer second opening outer perimeter portion extends to opposed outer perimeter portion sixth and eighth ends, and wherein a third curved edge of the washer second opening meets the fifth and sixth ends of the respective inner arcuate and outer perimeter portions at fifth and sixth junctions, and a fourth curved edge of the washer second opening meets the seventh and eighth ends of the respective inner arcuate and outer perimeter portions at seventh and eighth junctions, c) wherein, in an imaginary configuration having the washer first and second openings superimposed one on top of the other and when measured with respect to their orientations in the ceramic washer, the cumulative arcuate distance of the respective washer first and second opening inner arcuate portions, subtracting overlap, results in a gap between the superimposed washer first and second opening inner arcuate portions that is less than a distance between either of the first and third junctions or the fifth and seventh junctions.

2. The ceramic washer of claim 1, selected from Al.sub.2O.sub.3, BaO, CaO, CeO.sub.2, MgO, ZrO.sub.2, SiO.sub.2, TiO.sub.2, Al.sub.2SiO.sub.5, BaTiO.sub.3, SrTiO.sub.2, ZnO, Si.sub.3N.sub.4, AlN, BN, zirconia toughened alumina (ZTA), alumina toughened zirconia (ATZ), yttrium stabilized zirconia (YSZ), yttrium-toughened zirconia (YTZP), aluminum nitride (AlN), silicon nitride (Si.sub.3N.sub.4), boron nitride (BN), carbon nitride (CN), and combinations thereof.

3. The ceramic washer of claim 1, wherein the washer first and second inner arcuate portions of the respective washer first and second openings each extend for about 90° to about 180° to the first and third junctions for the washer first opening inner arcuate portion and to the fifth and seventh junctions for the washer second opening inner arcuate portion, and wherein the washer first and second outer peripheral portions of the respective washer first and second openings each extend for about 90° to about 180° to the second and fourth junctions for the washer first opening outer perimeter portion and to the sixth and eighth junctions for the washer second opening outer perimeter portion.

4. A ceramic washer, comprising: a) a first opening comprising at least two spaced-apart first opening inner arcuate portions that extend inwardly from a first opening outer perimeter comprising at least two spaced-apart first opening outer perimeter portions; and b) a second opening comprising at least two spaced-apart second opening inner arcuate portions that extend inwardly from a second opening outer perimeter comprising at least two spaced-apart second

opening outer perimeter portions.

5. The ceramic washer of claim 4, wherein at least the first opening comprises four spaced-apart first opening inner arcuate portions that extend inwardly from the first opening outer perimeter comprising at least four spaced-apart first opening outer perimeter portions.

6. The ceramic washer of claim 4, wherein the at least two spaced-apart first opening inner arcuate portions are diametrically opposed to each other.

7. The ceramic washer of claim 4, wherein at least one of the at least two spaced-apart first opening inner arcuate portions and the at least two spaced-apart second opening inner arcuate portions are diametrically opposed to each other.

8. The ceramic washer of claim 4, selected from Al_2O_3 , BaO , CaO , CeO_2 , MgO , ZrO_2 , SiO_2 , TiO_2 , Al_2SiO_5 , BaTiO_3 , SrTiO_2 , ZnO , Si_3N_4 , AlN , BN , zirconia toughened alumina (ZTA), alumina toughened zirconia (ATZ), yttrium stabilized zirconia (YSZ), yttrium-toughened zirconia (YTZP), aluminum nitride (AlN), silicon nitride (Si_3N_4), boron nitride (BN), carbon nitride (CN), and combinations thereof.

9. The ceramic washer of claim 4, wherein a first imaginary line centered between the at least two spaced-apart first opening inner arcuate portions is aligned at 12 o'clock to 6 o'clock, and a second imaginary line centered between the at least two spaced-apart second opening inner arcuate portions is aligned at 3 o'clock to 9 o'clock.
